Earth and Planetary Science Letters, 39 (1978) 199–210
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PALAEOMAGNETISM AND TECTONICS IN UMBRIA, ITALY

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Received August 12, 1977
Revised version received December 21, 1977

Detailed records of palaeomagnetic directions from two Upper Cretaceous Umbrian sections are compared with the composite section compiled by Van den Berg et al. [1]. In addition to these sections, fifty-one geographically distributed sites have been studied magnetically and palaeontologically. The Late Miocene–Pliocene fold belt of the Umbrian Apennines is arcuate in outcrop, being convex toward the east. The fold axes are generally tangential to the Umbrian arc, which can be divided into a northern northwestern-trending part and a southern northeastern-trending part. Sites from the north have more westerly declinations than those from the south. The differences are significant at the 95% probability level and most probably reflect the bending of originally straight fold axes or the development of fold axes with initial curvature. The palaeomagnetic directions from Umbria provide important information concerning the Umbrian tectonic deformation, but are not applicable to the Italian autochthon.

1. Introduction

The recent development of highly sensitive rock magnetometers has meant that the weakly magnetised pelagic limestones which characterise the basinal facies throughout the Periadriatic region can now be measured with sufficient precision. Apart from the Iblean zone of Sicily (fig. 1) where Cretaceous basalts have been studied palaeomagnetically [2–4], truly autochthonous units in Italy consists almost entirely of these weakly magnetised limestones. The Mesozoic South Tethyan limestones have been studied magnetically in three localities which have been considered autochthonous: Umbria, Apulia/Gargano and the Southern Alps (fig. 1). The Northern Apennine Umbrian sequence has recently been a focus of attention for palaeomagnetic workers in Italy. The sequence contains a pink pelagic limestone – the Scaglia Rossa – which represents almost continuous deposition from the Early Turonian to the Early Eocene. Several independent palaeomagnetic investigations of these basinal facies have produced mutually compatible data, but conflicting interpretations have arisen because of differing assumptions concerning the tectonic history of Umbria.

Lowrie and Alvarez [5,6] studied Scaglia Rossa samples from twenty localities distributed throughout Umbria. They interpreted the northwestward declinations as evidence for counterclockwise-rotation of Umbria and autochthonous peninsular Italy since the time of deposition of the Scaglia Rossa. They also placed sixteen sites at different stratigraphic levels in a complete section of the Scaglia Rossa near Gubbio, where the entire thickness of the formation from Turonian

Institut für Geophysik, ETH Zürich, Contribution Nr. 193.

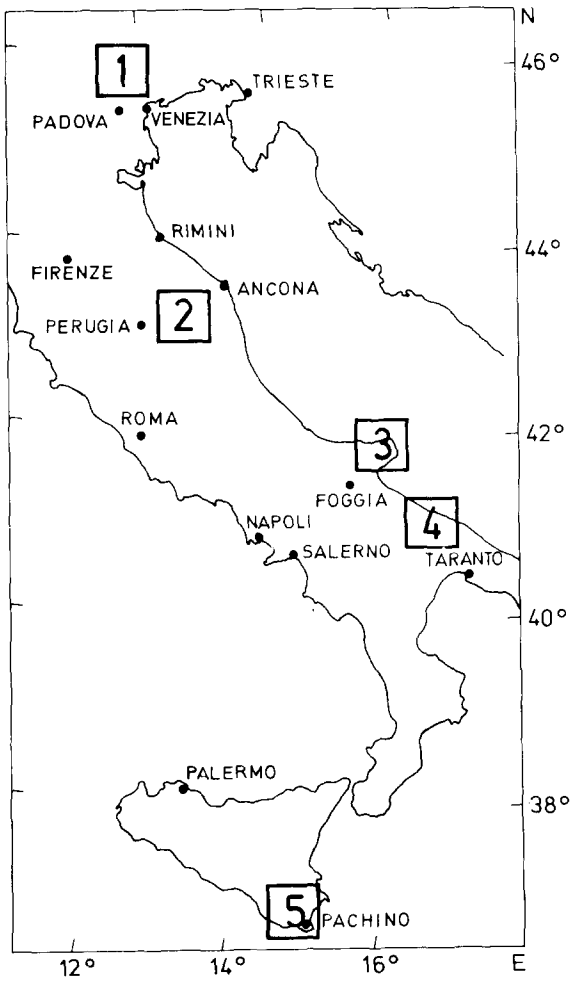


Fig. 1. Approximate location of the main regions which have been thought to be autochthonous and have been sampled for palaeomagnetic studies. 1 = Vicentinian and Venetian (Southern) Alps; 2 = Umbria; 3 = Gargano; 4 = Apulia; 5 = Iblei.

to Eocene is almost continuously exposed. Their data from this section showed a strong variation of declination with stratigraphic position which they interpreted as evidence for approximately 45° counterclockwise rotation of peninsular Italy during the Campanian-Maastrichtian followed by a further 25° counterclockwise rotation after the Middle Eocene. Lowrie and Alvarez [6] cautioned that this interpretation required confirmation from a check section.

Channell and Tarling [7] noted that the Late Cretaceous magnetic directions from Apulia, the Southern Alps and Iblei, were similar to those that would be expected if Italy moved as part of Africa since the Late Cretaceous. Their palaeomagnetic data from the Umbrian sequence were very similar to those of Lowrie and Alvarez and had declinations that were more westward than those associated with the other three "autochthonous" palaeomagnetic sampling localities. They suggested that the Umbrian sequence directions were the result either of allochthony or of decoupling of the Northern Apennine "autochthon" from the rest of autochthonous Italy along the Anzio-Ancona line and beneath the Po basin.

Full interpretation of the regional data of both these investigations was impeded by absence of palaeontological dates for the sites. This is essential as deposition of the Scaglia Rossa alone spanned an interval of about 50 m.y., from Early Turonian to Middle Eocene.

Van den Berg et al. [1] sampled several stratigraphic levels within the Umbrian sequence. They recognised that the shifts of magnetic direction with time were similar to those expected for Africa, but that the whole pattern was rotated by 25–30° westwards. Van den Berg and Wonders [8] pointed out that this "African pattern" does not coincide with a similar but unrotated pattern revealed by sampling the same stratigraphic levels in the Vicentinian (Southern) Alps and proposed that the data can be explained if autochthonous Italy, south of the Po basin, moved with Africa until post-Early Eocene time when a 25–30° counterclockwise rotation of the Italian peninsula occurred relative to Africa. The Southern Alps were considered to have been detached from autochthonous Italy during the Late Cretaceous and to have undergone a clockwise rotation relative to Africa and a later counterclockwise rotation of the same magnitude such that there is now consistency between Southern Alpine and African data. This interpretation assumes that the Umbrian sequence is autochthonous and that the palaeomagnetic data from Umbria are applicable to peninsular Italy. An opposing view advocating allochthony in Umbria was presented by Channell [15].

The different conclusions result in large part from the differences of geological opinion concerning the autochthony of Umbria. The present investigation was undertaken as a contribution to the resolution of this problem, by (1) bringing together the data from the separate stratigraphic sections under a common interpretation, and (2) combining data from all 51 palaeomagnetic sites geographically distributed throughout Umbria, each of which has now been dated palaeontologically.

2. Palaeomagnetic results from stratigraphic sections in Umbria

The Upper Cretaceous section at Gubbio and a check section at Moria (Fig. 2) have been intensively sampled for the purpose of establishing a Late Cretaceous magnetic stratigraphy. At Gubbio over 600 samples were collected at approximately 0.5-m intervals [9,10] and at Moria over 250 samples were collected at approximately 1-m intervals throughout the sections. The two sections have produced well-defined magnetic stratigraphies which can be correlated with each other and with oceanic magnetic anomalies. The Gubbio section has been dated palaeontologically by Premoli-Silva [11]. The Moria section has been dated by correlating both the reversal boundaries and the recognisable litho- and biostratigraphic markers (the 1-m Turonian bituminous shale bed known as the "Bonarelli level", the top of the chert-bearing lower member of the Scaglia Rossa, and the Cretaceous-Tertiary boundary) with the section at Gubbio. The correlation is almost exactly linear and establishes that the two sections had more or less identical sedimentation rates. The individual sample directions have been averaged over 10-m stratigraphic intervals; the declinations and inclinations of these "site" means, together with error bars representing the 95% confidence ranges, are shown for the Gubbio section in Fig. 3 and for the Moria section in Fig. 4.

The Umbrian palaeomagnetic data of Van den Berg et al. [1] came from three sections located near the town of Cagli (Fig. 2); one of these was also the Moria section. About ten sites were sampled at each section.

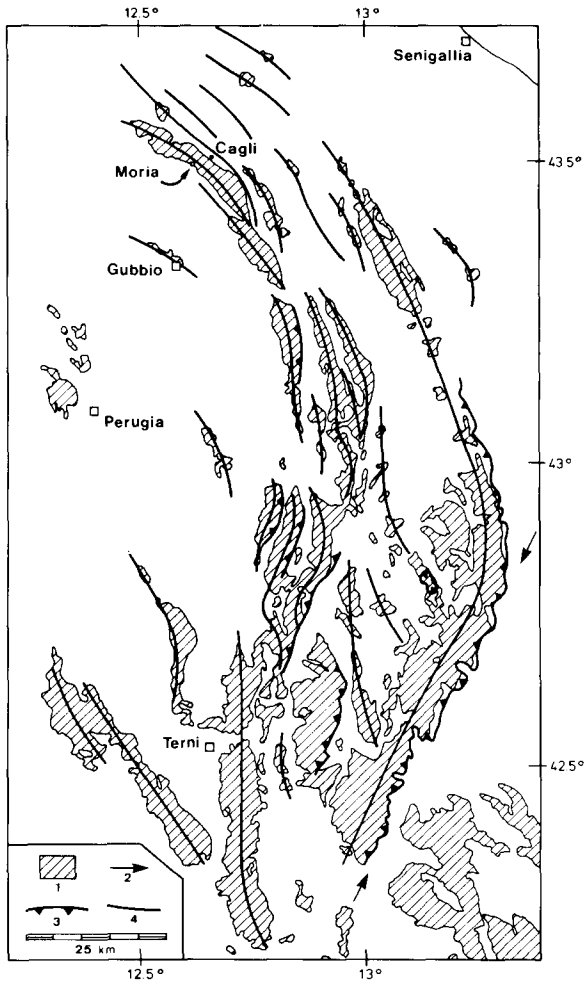


Fig. 2. Tectonic sketch map of the Umbrian Apennines. 1 = outcrop areas of the lower part of the stratigraphic section (Upper Triassic–Lower Cretaceous); 2 = Anzio-Ancona Line; 3 = major reverse faults or thrusts; 4 = anticlinal axes.

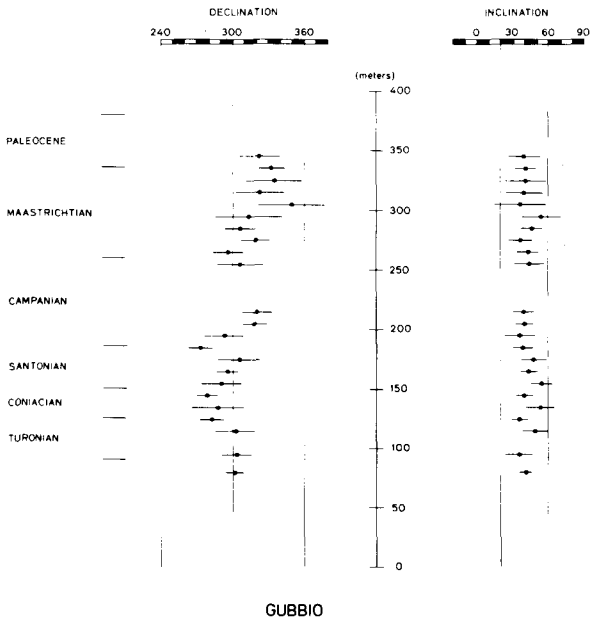


Fig. 3. Progressive changes in declination and inclination with age in the Gubbio section. The values are obtained by averaging the sample directions over 10-m stratigraphic intervals. The error bars correspond to α_{95} for the inclination values and $\alpha_{95}/\cos I$ for the declination values.

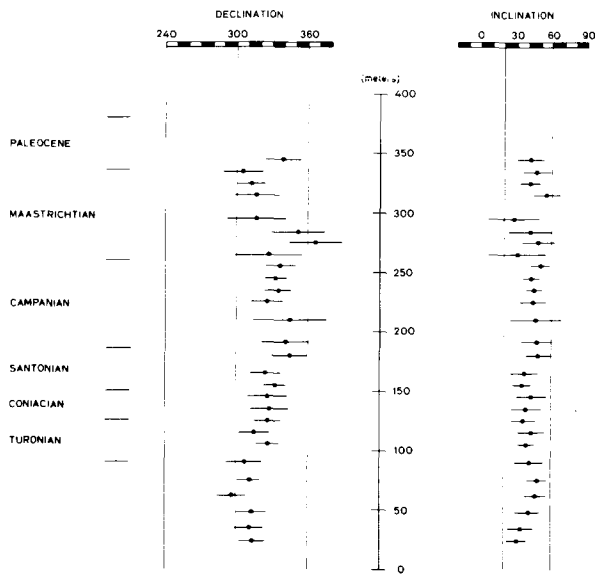


Fig. 4. Progressive changes in declination and inclination with age in the Moria section averaged over 10-m intervals. The vertical scale corresponds to the sedimentation rate at Gubbio and Moria. The error bars are obtained as in Fig. 3.

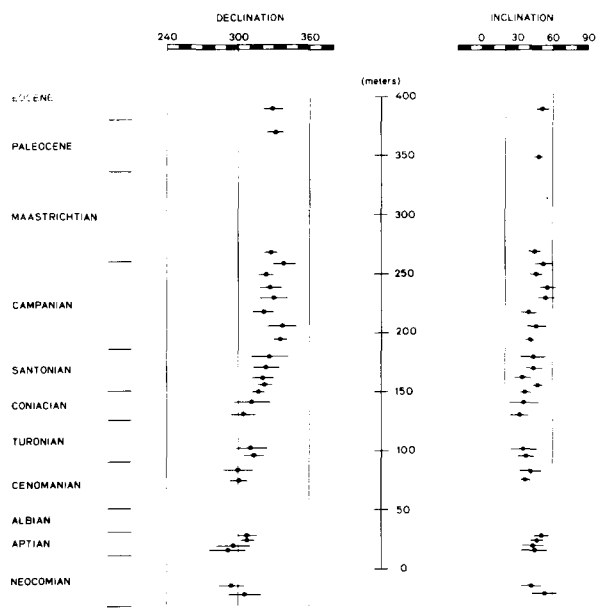


Fig. 5. Progressive changes in declination and inclination with age in the composite section derived by Van den Berg et al. [1]. The vertical scale corresponds to the sedimentation rate in the Gubbio and Moria sections. The error bars are obtained as in Fig. 3.

The data were combined on the basis of palaeontological dating to give a composite record of the progressive changes in declination and inclination during the Cretaceous. The locations of the stage boundaries were interpolated from the Gubbio to the Moria section and were then correlated with the corresponding stage boundaries of Van den Berg et al. [1] in order to plot the data of the composite Cagli section on the same scale as the other two sections (Fig. 5). A well-defined increase in declination is recognised during the Coniacian to Santonian interval, accompanied by a less well-defined decrease and then increase in inclination during the Cenomanian to Coniacian interval. These directional changes were considered by Van den Berg et al. to constitute an "African pattern" because they produce a polar wander curve similar to that inferred for Africa by Van der Voo and French [12].

The scatter in each of the three sets of declinations (Figs. 3, 4, 5) makes immediate comparisons difficult, therefore generalized trends in declination were derived for each section (Fig. 6). The upper confidence limits were passed through a 5-point, weighted moving filter and a smooth curve was drawn through the ensuing points. A similar smoothed curve was produced for the lower confidence limits. The zone between these bounds represents the declination trend for the section.

On the basis of these declination trends (Fig. 6) and the data of the individual sections (Figs. 3, 4, 5) the following observations may be made:

(1) Because the Moria section constitutes one of the three sections sampled by Van den Berg et al. [1] close consistency is expected between their composite Cagli section and the Moria section. The directions from these two sections are indistinguishable and the increase in declination during the Turonian-Santonian is apparent in both sections (Fig. 6), but the accompanying decrease, then increase, in inclination in the Cagli section (Fig. 5) is not really apparent in the data from the Moria section (Fig. 4).

(2) The inclination data from Gubbio are consistent with those from Moria and likewise show no well-defined change throughout the Late Cretaceous. Comparison of the declination trends indicates an increase

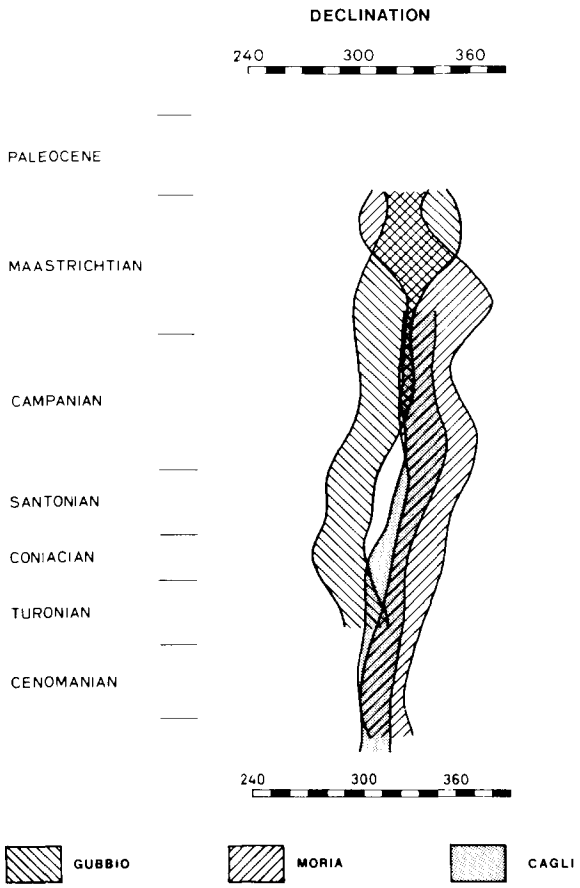


Fig. 6. The envelope of the declination error bars for each section smoothed using a 5-point moving filter.

in declination in the Gubbio section beginning in the Coniacian, which appears to parallel the increase in declination in the same time interval in the Moria section and in the composite Cagli section (Fig. 6). Statistical tests [13,14] of the Gubbio and Moria directions show that they are indistinguishable after the middle Maastrichtian. The errors are largest in the Maastrichtian in both sections.

The uniformity of lithology and sedimentation rate between the Gubbio and Moria sections and their geographical proximity indicate that these limestones were deposited in the same part of the basin. Taken in conjunction with the similar characteristics of the remanent magnetisations and the correlation of the two magnetic stratigraphies, this implies that the differences between the Gubbio and the Moria/Cagli declinations do *not* result from differences in magnetic properties.

The Gubbio and Moria directions respond positively to the tectonic fold test [16], indicating that the palaeomagnetic directions predate the tectonic folding in Umbria. The average inclinations in the two sections are very nearly the same after bedding corrections, assuming horizontal fold axes, this assumption is probably not greatly in error. The parallelism between the Gubbio and Moria/Cagli pre-Maastrichtian declination trends (Fig. 6) suggests that differences in declinations in the two sections may be partly the result of a relative rotation of fold axes of some 20° in the two regions about a sub-vertical axis.

Although haematite is a common magnetic component in the Umbrian Scaglia Rossa, it usually carries the same direction as the more dominant primary magnetite component, and for this reason is thought to have been produced soon after deposition [6,10]. However, the agreement of directions and the increased size of the error bars in the Maastrichtian in the different sections (Fig. 3–6) may reflect the presence of a secondary overprint associated with later hematization.

3. Regionally distributed sites in Umbria

The combined data of previous studies [5,7] were enlarged by adding six new sites. This enlarged data base comprising 51 sites, located throughout Umbria, is an improvement over earlier studies. The dependence of the palaeomagnetic directions upon geographical position can now be studied, and the data base is large enough to permit statistical analysis.

The determination of fold axes at each site was not possible, but their general geometric arrangement was obtained from recently compiled geological maps. The fold axes are approximately tangential to the arcuate outcrop of the folded Umbrian Apennines (Fig. 2), and the site mean declinations appear to vary systematically around the Umbrian arc (Fig. 7). This variation could result from two effects: (1) tectonic rotations about sub-vertical axes or (2) variation of age of the sites. In order to understand the second of these effects each of the sites has been dated palaeontologically.

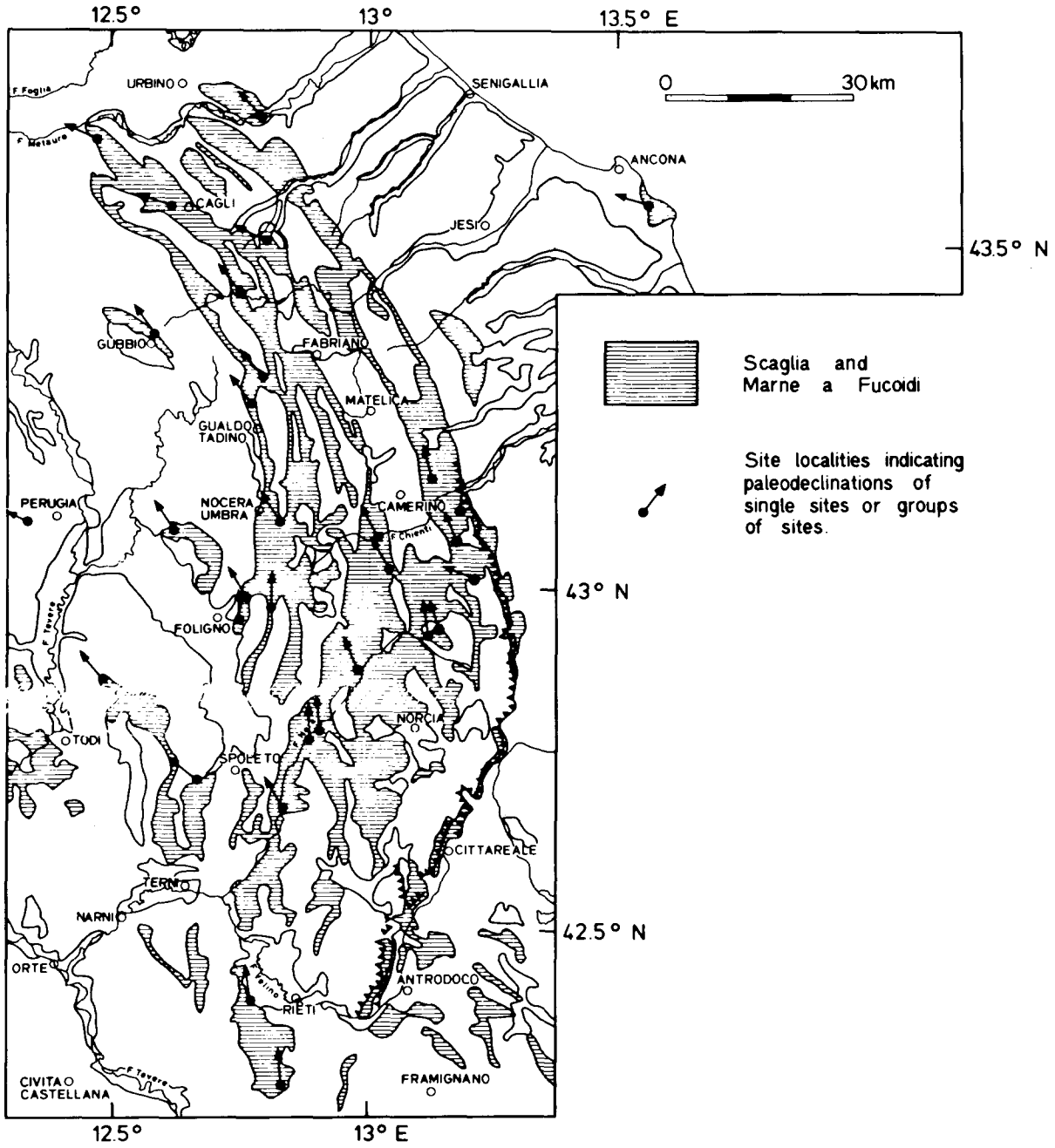


Fig. 7. The location of the geographically distributed palaeomagnetic sites. Arrows indicate the palaeodeclinations of individual sites or groups of sites.

3.1. Biostratigraphy

The Middle Cretaceous to Lower Tertiary pelagic sediments of the Umbrian sequence contain a succession of planktic foraminiferal faunas and nannofossil assemblages similar to those from other areas of the Tethyan realm. Several biostratigraphic studies by means of planktic foraminifera have been made on this

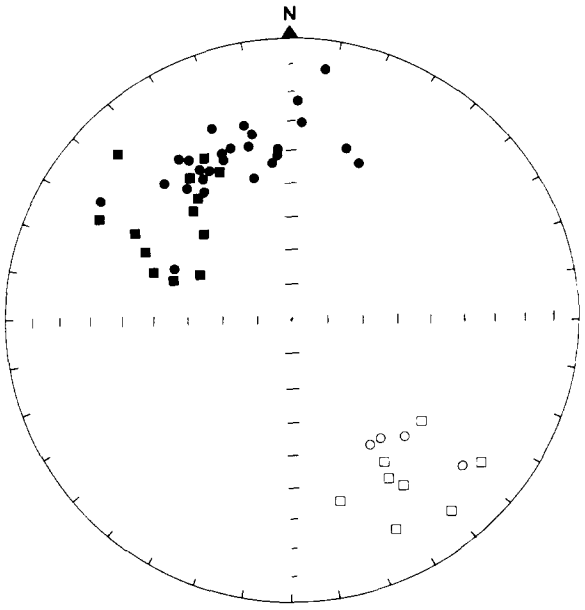


Fig. 8. Stereographic plot of the geographically distributed site mean directions. Circles: south Umbrian sites. Squares: north Umbrian sites. Closed symbols: lower hemisphere. Open symbols: upper hemisphere.

sequence, in particular on the classic Gubbio section [11, 17–20]. As a result, a detailed biostratigraphic scheme based on planktic foraminifera is now available for the late Santonian to Eocene interval in this region. The only published data on the calcareous nannofossils of the Umbrian sequence are those of Monechi and Pirini Radrizzani [21] who described some assemblages at the Cretaceous-Tertiary boundary using electron microscopy.

In order to obtain as accurate palaeontological dating of the geographically distributed sites as possible, both planktic foraminifera and calcareous nannofossils have been studied. The biostratigraphic study by means of planktic foraminifera was carried out on thin sections or dry peels prepared from the same core samples taken for palaeomagnetic analysis. Due to the general abundance of microfossils, the use of several thin sections or dry peels for each sample made it possible to find zonal markers and to obtain a fairly good knowledge of the compositions of the foraminiferal assemblages. Thus it was possible to identify for each assemblage the biostratigraphic zone to which it belonged. For only a few Maastrichtian samples it was difficult to establish with certainty – by means of sections alone – whether they should be assigned to the *Globotruncana tricarinata* Zone or *Globotruncana gansseri* Zone. This is due to the rarity of the zonal marker *G. gansseri* in the homonymous zone of the Umbrian sequence and the consequent difficulty of finding it even in several thin sections. In these cases a more precise biostratigraphic dating was obtained by means of calcareous nannofossils contained in the same samples.

The planktic foraminifera zonal scheme used in this paper for age determinations is shown in Fig. 9. The zonation of the Coniacian to Eocene interval is mainly according to Premoli Silva and Bolli [22] and Premoli Silva [11]; that of the Albian to Turonian is according to Van Hinte [23].

The same samples which were taken for palaeomagnetic study were also investigated for their nannofossil content. The assemblages were poorly preserved because of diagenetic changes. However, in the Campanian to Maastrichtian samples all the zonal markers, as well as most of the other important species, were still identifiable with an optical microscope. Therefore, for this stratigraphic interval the assemblages could be correlated to the nannofossil zonal scheme shown in Fig. 9. A definition of these nannofossil zones is given in papers by Martini [24] and Proto Decima et al. [25].

Nannofossils from the two Tertiary sites were so highly altered by diagenetic changes that it was nearly always impossible to ascertain their specific identities.

3.2. Palaeomagnetic directions at the palaeontologically dated sites

The results of the palaeontological study are combined with the palaeomagnetic data in Fig. 10. The length of the vertical bar for each site indicates the maximum uncertainty in its stratigraphic position, based on the widths of the stratigraphic ranges of both planktic foraminifera and nannofossil assemblages. The error in declination ($\alpha_{95}/\cos I$) at each site is given by the width of the horizontal error bar; inclination error bars have been omitted to avoid cluttering the figure. Within individual palaeontological zones, there is a large spread especially in declination.

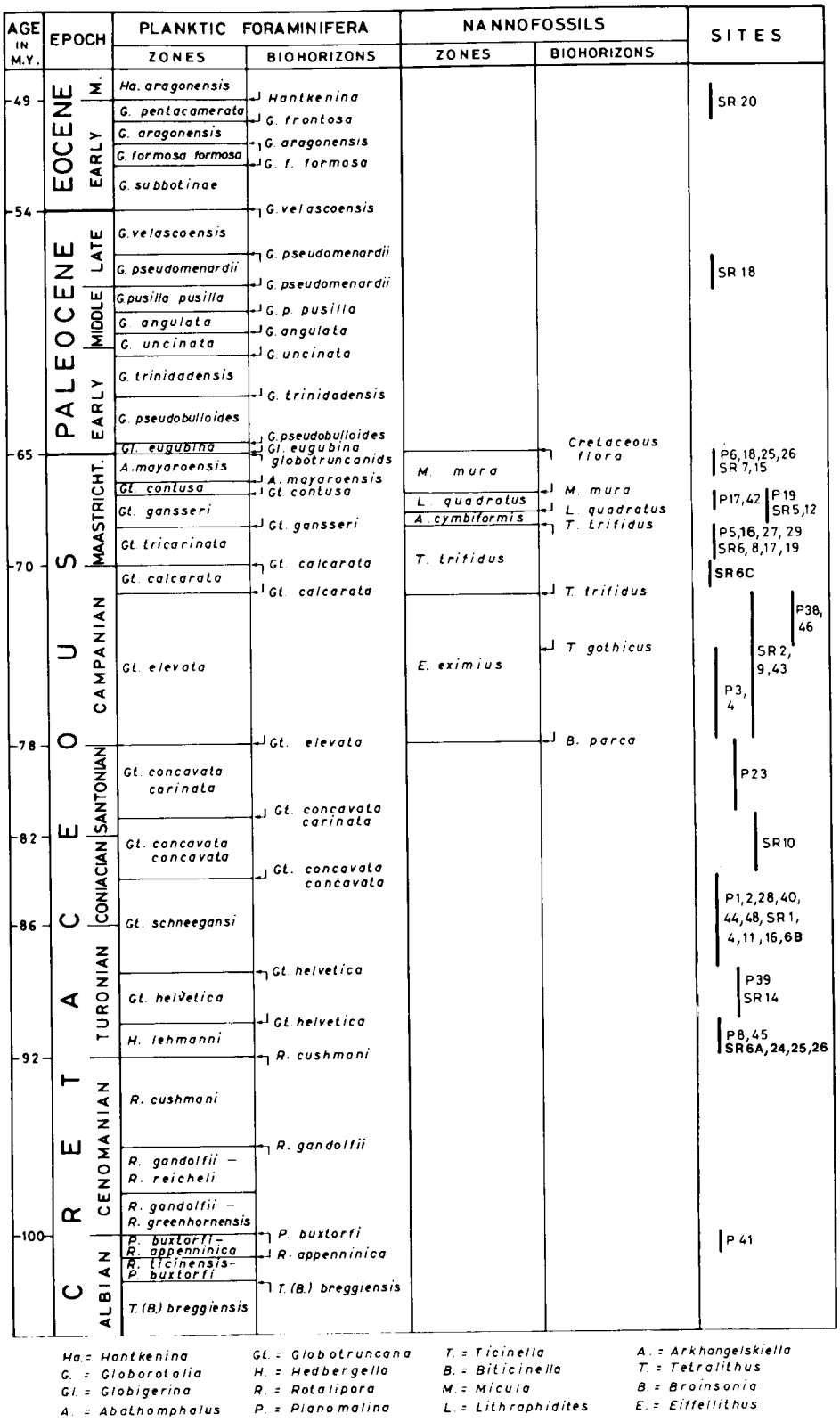


Fig. 9. The biostratigraphic position of the geographically distributed sites based on planktic foraminifera and nannofossils.

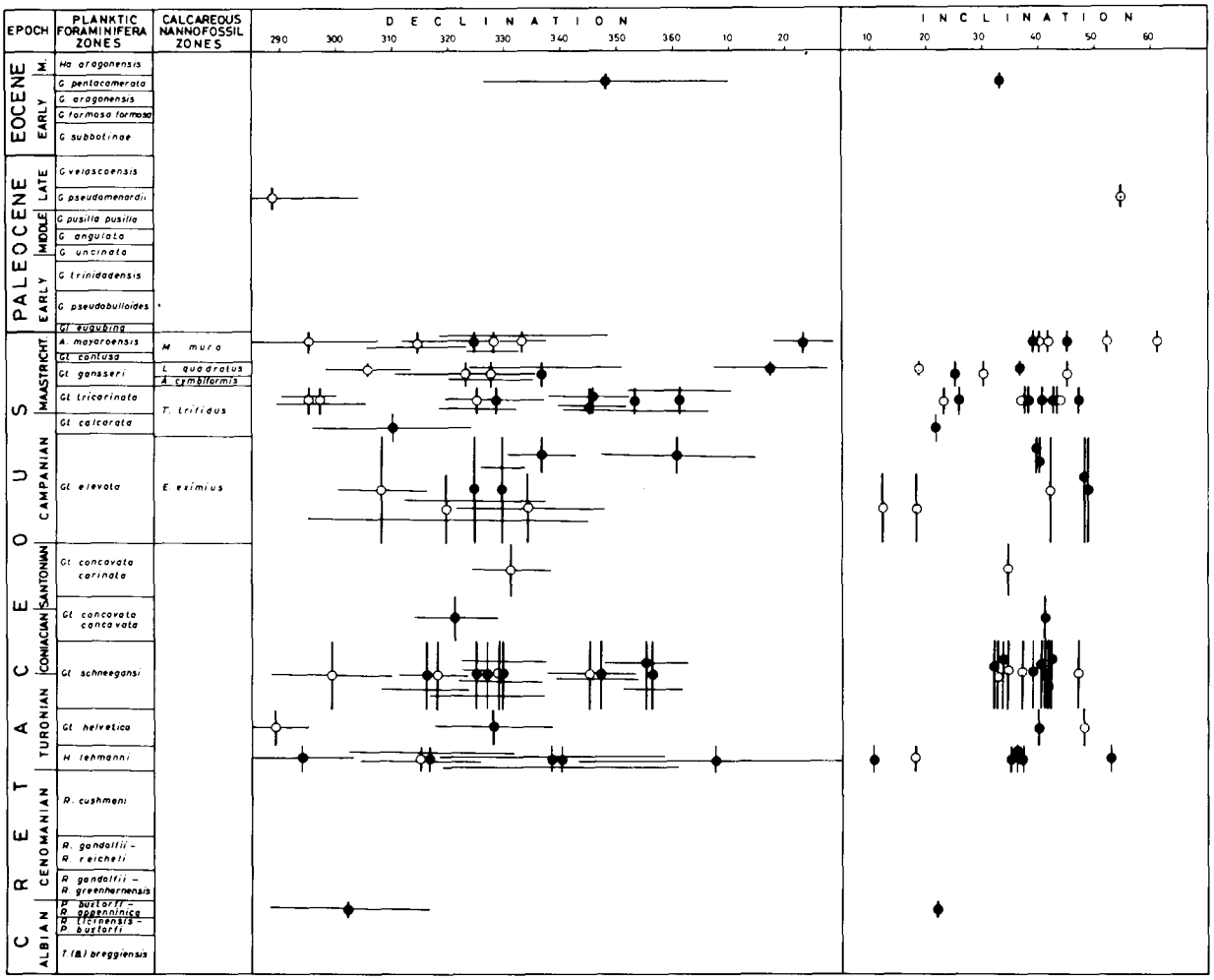


Fig. 10. Site mean declinations and inclinations plotted as a function of stratigraphic position. The open points represent sites from the northern half of Umbria and the closed points represents those from the south. The vertical bars represent the possible stratigraphic range of the sites and the horizontal bars represent the error in declination ($\alpha_{95}/\cos I$).

At least part of the scatter may be due to local tectonics.

Generally speaking, in the northern part of the Umbrian Apennines the fold axes trend northwest-southeast. In the southern part the fold axes orientations are more variable, on the eastern limb they trend northeast-southwest. The difference in style is gradational and occurs roughly at the latitude of Perugia (Fig. 2).

We have used this arbitrary division to divide our sites into a "northern" group of 21 sites and a "southern" group of 30 sites (Figs. 8, 10). The inclinations do not appear to be different in the two groups, but the declinations show a marked difference. Those from the northern group appear to have more westerly declinations than those from the southern group. It is not possible to test statistically the significance of this difference for each palaeontological zone, as insufficient sites are usually available to make a meaningful statistical test. However, the majority of sites occur in two time intervals: the Maastrichtian, and the Turonian/Coniacian. The results of an F-test [14] indicate that the mean directions of the "northern" and "southern" groups are significantly different at

TABLE 1

Statistical comparison of site mean directions from north and south Umbria *

Description	Number of sites	Mean decl.	Mean incl.	α_{95}	R	k	F-ratio	Null hypothesis
Maastrichtian:								
North group	10	315	41	10.3	9.605	22.8	11.44	3.30
South group	9	351	39	11.5	8.618	21.0		
Turonian and Coniacian:								
North group	6	317	38	16.5	5.712	17.3	2.48	3.30
South group	13	335	38	9.2	12.44	21.4		
All sites, Albian to Eocene:								
North group	21	316	40	7.3	19.99	19.8	9.75	3.10
South group	30	338	39	6.2	28.47	19.0		

* R is the resultant vector, k the precision parameter, and α_{95} the radius of the circle of 95% confidence [13]. The significance of the difference in direction between north and south groups is tested by comparing an F-ratio [14] against its value for the null hypothesis.

the 95% confidence level for the Maastrichtian (Table 1), the difference being primarily due to declination differences. The difference in the Turonian/Coniacian is not statistically significant, probably because of the large error associated with the small number of northern sites. When all sites are considered the difference between north and south is again significant.

4. Discussion and conclusions

The declinations in the intensively sampled sections (Figs. 3–6) display systematic trends in which the change of declination with stratigraphic position at a single locality is of the same magnitude as the regional declination difference between northern and southern Umbria (Table 1). Therefore, as previously acknowledged [6], a complete analysis of the regional declination pattern is impossible without compensating for the declinations' variation with age. Palaeontological dating of all regionally distributed sites demonstrates that there is no statistically significant dependence of the declinations upon age of the site (Fig. 10). However, significant directional differences between the northern and southern groups exist, even within a single stratigraphic stage (Table 1). The regional declination pattern (Fig. 7) therefore results from the Tertiary tectonism that affected the Umbrian Apennines.

The progressive change in declination from north to south and the difference between the north and south regional mean palaeomagnetic directions (Table 1) are probably the result of the variable amount and nature of the tectonic deformation in different parts of the mountain belt. The tectonic styles in the north and south are not the same. In the northern part the Umbrian arc is formed by fairly simple, subparallel folds, whereas the southern part is more complex, with the development of major reverse faults or thrusts. This may indicate that the fold belt resulted from movements along the Anzio-Ancona line, with the effects dying out toward the north (B. D'Argenio and G. Pialli, personal communitation, 1977).

The arcuate arrangement of the fold axes, and the generally gradational declination change, may have been achieved by either (1) the generation of fold axes with original curvature, or (2) the bending of a pre-existing fold system (oroclinal bending). A Triassic evaporitic unit, known from wells and occasional outcrops, could have facilitated décollement of the upper part of the section and permitted the movements required by either mechanism.

The palaeomagnetic directions also show a more local tectonic influence; the local deviations of directions from the regional means (typified in the north by the differences between the Gubbio and Moria sections) are occasionally comparable to the difference between the regional means. This is probably due to local differences in the amounts of fold axes rotations.

The palaeomagnetic data do not in themselves provide a definitive solution as to the nature of the regional or local tectonic deformation but provide information to supplement field studies which are currently in progress.

The European polar wander path of Van der Voo and French [12] predicts a declination/inclination combination for the Late Cretaceous in Umbria of 357°/44°, while their African polar wander path predicts a direction of 342°/36°. The mean direction of our sites in the northern half of Umbria is 316°/40°, and for the sites in the southern half is 338°/39° (Table 1). "European" directions are unlike those in either the northern or southern half of Umbria. The directions from the sites in the southern half of Umbria agree quite closely with "African" directions; however, the mean declination of the northern group has been rotated some 22° counterclockwise relative to the southern group and to "African" directions.

Too much significance should not be placed upon these "African" directions from southern Umbria. The tectonic situation in Umbria is as yet incompletely understood; considerable transport of the sedimentary cover may have occurred. Also the Late Cretaceous palaeomagnetic pole positions for both Europe and Africa are not well defined. Furthermore, although our demarcation between northern and southern groups appears to correspond to a difference in tectonic styles and fold axes trends, the transition is gradational and there is no tectonic feature with which the demarcation can be identified. Thus the "African" directions in southern Umbria may well be fortuitous.

Channell and Tarling [7] recognised a westward deviation of Umbrian data relative to Southern Alpine and African data of the same age at sites distributed mainly in the northern part of Umbria. Van den Berg and Wonders [8] interpreted similar differences in palaeomagnetic data from the Southern Alps and Umbria in terms of relative rotation of two autochthons bounded by a fault in the Po basin. Their Umbrian data came from near Cagli, in the northern part of Umbria. Our investigation of dated sites indicates that the westward deviation of magnetic directions reflects the progressive increase in counterclockwise rotation of the northern half of Umbria relative both to the southern half of Umbria and to the Southern Alps. This casts doubt on the validity of Van den Berg and Wonders' interpretation. Palaeomagnetic data from Umbria yield information concerning the tectonic deformation of the Umbrian Apennines but are not applicable to the Italian autochthon.

Acknowledgements

We are grateful to our Dutch colleagues, J. Van den Berg, C.T. Klootwijk, and A.A.H. Wonders, who kindly gave us a preprint of their article [1] and who have allowed us to use these data in our palaeomagnetic analysis of different Umbrian sections in the present paper.

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